A PROPER MORPHISM WITH FINITE FIBRE OVER A ONE DIMENSIONAL POINT IS FINITE OVER A NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The list of hypotheses ends with the author's own placeholder 'add more here'; that item is reproduced verbatim and no condition was added, removed or reordered.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-finite-in-codim-1 is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Quasi-finiteness at points over a one dimensional local ring. Same chapter, Lemma lemma-quasi-finite-in-codim-1, invoked at proof lines 3289-3290. Supplies quasi-finiteness at every point of the fibre.
- Quasi-finiteness at a point makes the fibre discrete there. Morphisms, Lemma morphisms-lemma-quasi-finite-at-point-characterize, invoked at proof lines 3290-3292. Supplies discreteness of the fibre.
- A proper morphism with finite fibre over a point is finite over a neighbourhood. Cohomology of Schemes, Lemma coherent-lemma-proper-finite-fibre-finite-in-neighbourhood, invoked at proof lines 3293-3296. Supplies the neighbourhood on which the morphism is finite.

Source reference. Original lemma 02OH: sources/stacks-chow/tags.txt, source lines 3516–3516. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AB6: sources/stacks-varieties/pinned-original-source.tex, source lines 3188–3188. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TH: sources/stacks-chow/tags.txt, source lines 2374–2374. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism. Let $y \in Y$ be a point such that $\mathcal{O}_{Y,y}$ is Noetherian of dimension ≤ 1. Assume in addition one of the following conditions is satisfied*

(1) *for every generic point η of an irreducible component of X the field extension $\kappa(\eta)/\kappa(f(\eta))$ is finite (or algebraic),*
(2) *for every generic point η of an irreducible component of X such that $f(\eta) \rightsquigarrow y$ the field extension $\kappa(\eta)/\kappa(f(\eta))$ is finite (or algebraic),*
(3) *f is quasi-finite at every generic point of X,*
(4) *Y is locally Noetherian and f is quasi-finite at a dense set of points of X,*
(5) *add more here.*

Then there exists an open neighbourhood $V \subset Y$ of y such that $f^{-1}(V) \to V$ is finite.

Proof. By Lemma 0AB6 the morphism f is quasi-finite at every point of the fibre X_y. Hence X_y is a discrete topological space (Morphisms, Lemma 01TH). As f is proper the fibre X_y is quasi-compact, i.e., finite. Thus we can apply Cohomology of Schemes, Lemma 02OH to conclude. □